PHAS0042

Quantum Mechanics

Practice Paper 2022-23

[Browne/Monteiro]

Answer ALL THREE questions

The numbers in square brackets show the provisional allocation of maximum marks per question or part of question.

[Part marks]

1. The Hamiltonian $\hat{H}_0$ of a particle of mass m moving in a quadratic potential takes the form:

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2}k_1\hat{x}^2$$

   (a) Represent $\hat{H}_0$ in matrix form using its eigenstates $|n\rangle$ as a basis, including only the lowest three energy eigenstates and assuming the natural frequency of the oscillator is ω. **[3]**

   (b) At time $t = 0$, the system is prepared in its **first excited** state. However, subsequently the potential is subject to a sudden perturbation $\hat{H}_1 = k_2\hat{x}^2$ so that $\hat{H} = \hat{H}_0 + \hat{H}_1$.
   Obtain the matrix of the total Hamiltonian, in the basis of $\hat{H}_0$, retaining only the 3 lowest energy eigenstates. Show all your working. **[5]**

For the remainder of this question, consider the case where $k_1 = 2k_2$:

   (c) Calculate the eigenenergies of $\hat{H}$ using the matrix you obtained in (b). You may reorder the basis if it seems useful. **[6]**

   (d) Now, treating $\hat{H}_1$ as a perturbation, use first order perturbation theory to calculate the ground and first two excited state energies. **[6]**

[Part marks]

2. (a) Two particles have spin angular momenta $s_1 = 1/2$ and $s_2 = 1/2$ respectively. Give the form of all total angular momentum states $|SM\rangle$ showing how they may be constructed from the single particle states $|s_1 m_1\rangle$ and $|s_2 m_2\rangle$. **[3]**

   (b) Express the total spin operator $\hat{S}^2$ in terms of single particle spin operators $\hat{s}_k$, for $k = 1, 2$. Include raising and lowering operators $\hat{s}_{\pm k}$, in your expression. Show all your working. **[3]**

   (c) Consider the Hamiltonian $\hat{H} = A\hat{S}^2 + \beta\hat{S}_z^2$. Represent this Hamiltonian in the basis $|s_1 m_1\rangle|s_2 m_2\rangle$, carefully explaining your working. **[6]**

   (d) Obtain all the eigenvalues and eigenstates corresponding to this Hamiltonian. **[5]**

   (e) The system is now prepared in a spin state that is a superposition

$$|\chi\rangle = \sum_{S,M} C_{SM}|SM\rangle$$

   of all S, M that are physically allowed. For the case $C_{10} = \frac{1}{\sqrt{3}}$, $C_{00} = \frac{2}{\sqrt{3}}$, while all other $C_{SM} = 0$, calculate the expectation value of the Hamiltonian $\hat{H}$ given above. **[3]**

[Part marks]

3. (a) A potential well is created in the lab. If a single particle is in the well, the ground and first excited states have wave functions $\psi_1(\mathbf{r})$ and $\psi_2(\mathbf{r})$ with associated energies E_1 and E_2.

   i. Two non-interacting spin zero particles are placed in the the well. What is the wave function of the ground state of this system? Justify your answer. **[4]**

   ii. Two non-interacting spin half particles are placed in the well instead. They have been prepared in the joint spin state $(1/\sqrt{2})(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)$. Assuming that they remain in this spin state, what is the joint spatial wave function that will minimize their energy? Fully justify your answer. **[5]**

   (b) A single particle is trapped in a one-dimensional well with the potential: $V(x) = \infty$ when $x \leq 0$ and $V(x) = cx$ elsewhere. Consider a trial wave function $\psi_\alpha(x) = Axe^{-\alpha x}$, where $\alpha > 0$.

   i. Compute the value of A which normalises $\psi_\alpha(x)$. **[3]**

   ii. Show that the expectation value of potential energy for the normalised $\psi_\alpha(x)$ is $(3c)/(2\alpha)$. **[3]**

   iii. You are given that the expectation value of kinetic energy for the normalised $\psi_\alpha(x)$ is $(\hbar^2\alpha^2)/(2m)$. Using this and the results of bi) and bii), use the variational method with trial solution $\psi_\alpha(x)$ over parameter α to derive an estimate of the ground state energy of the well of the form $E \approx \beta\,(c^2\hbar^2/2m)^{1/3}$, where β is a number. Write β as a decimal to 3 s.f.. **[5]**

   You may make use of the following integrals:

$$\int_0^\infty xe^{-2\gamma x}dx = \frac{1}{4\gamma^2} \qquad \int_0^\infty x^2e^{-2\gamma x}dx = \frac{1}{4\gamma^3} \qquad \int_0^\infty x^3e^{-2\gamma x}dx = \frac{3}{8\gamma^4}$$